GCSE BIOLOGY

Higher Tier – Paper 1 style

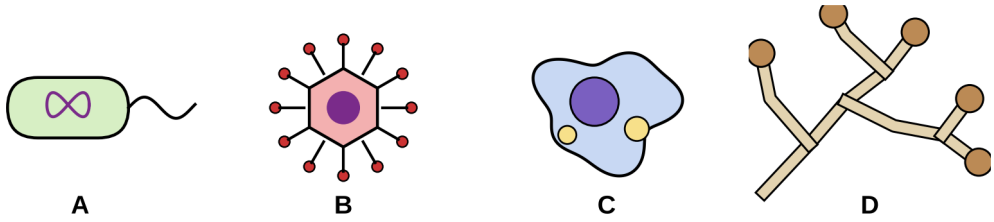
Topic 4.3: Infection and response

Practice questions in the style of AQA GCSE Biology (8461). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 190 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a ruler, a pencil and a calculator.
Instructions	• Use black ink or a black ball-point pen (pencil for diagrams and graphs). • Answer **all** questions. • Answer the questions in the spaces provided. • Do all rough work on a separate sheet. • Cross through any work you do not want to be marked.
Information	• The maximum mark for this paper is **180**. • The marks for questions are shown in brackets. • You are allowed to use a calculator. • You are reminded of the need for good English and clear presentation in your answers. • Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence.

Name: _____ Date: _____ Mark: _____ / 180

0 1 Pathogens are microorganisms that cause infectious disease. The diagrams show four different pathogens, **A** to **D**.



0 1 . 1 What is meant by a **pathogen**? **[1 mark]**

0 1 . 2 Which diagram shows a **virus**? Choose a letter. **[1 mark]**

0 1 . 3 Which diagram shows a **fungus**? Choose a letter. **[1 mark]**

0 1 . 4 Pathogen **C** is a protist. Name a disease caused by a protist. **[1 mark]**

0 1 . 5 Describe how bacteria make people ill. **[2 marks]**

0 1 . 6 Describe how viruses make people ill. **[2 marks]**

0 1 . 7 Pathogen **A** can divide every 20 minutes. Calculate the number of pathogens produced from **one** pathogen after 3 hours. Assume none die. **[3 marks]**

Answer = _____

0 2 The table gives information about some communicable diseases.

0 2 . 1 Complete the table. **[5 marks]**

Disease	Type of pathogen	How it is spread
Measles		droplets from coughs and sneezes
Salmonella food poisoning		contaminated food
Malaria		
Rose black spot		spores carried by wind and water
HIV	virus	sexual contact / exchange of body fluids

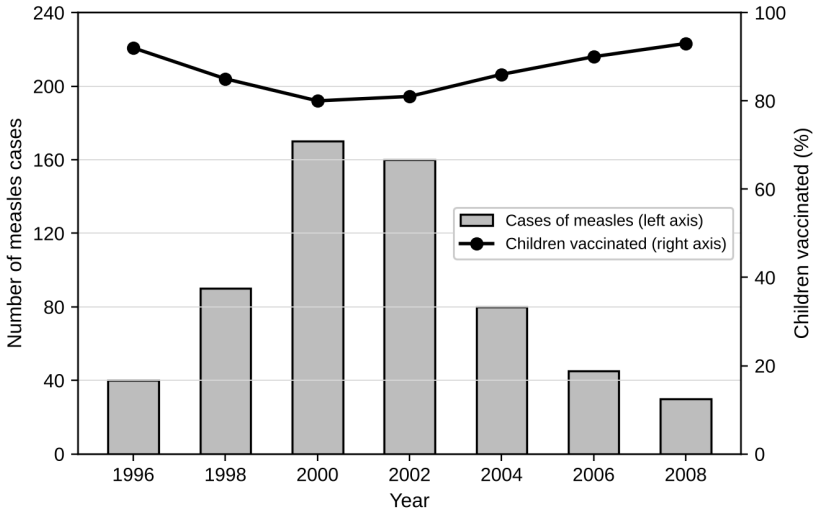
0 2 . 2 Describe **two** ways in which the spread of Salmonella food poisoning can be reduced. **[2 marks]**

0 2 . 3 Explain how washing hands can reduce the spread of disease. **[2 marks]**

0 2 . 4 Name **one** symptom of measles and **one** symptom of HIV infection. **[2 marks]**

0 2 . 5 Measles can be fatal. Explain why measles is spread so easily in a school. **[2 marks]**

0 3 A city had an outbreak of measles. Some parents stopped vaccinating their children after a news story about the vaccine. The graph shows the number of cases of measles and the percentage of children who had been vaccinated between 1996 and 2008.



0 3 . 1 State the number of cases of measles in 1998 and in 2000. **[2 marks]**

0 3 . 2 Calculate the percentage increase in the number of cases of measles between 1998 and 2000. **[3 marks]**

Answer = _____ %

0 3 . 3 Describe the relationship between the percentage of children vaccinated and the number of cases of measles. **[2 marks]**

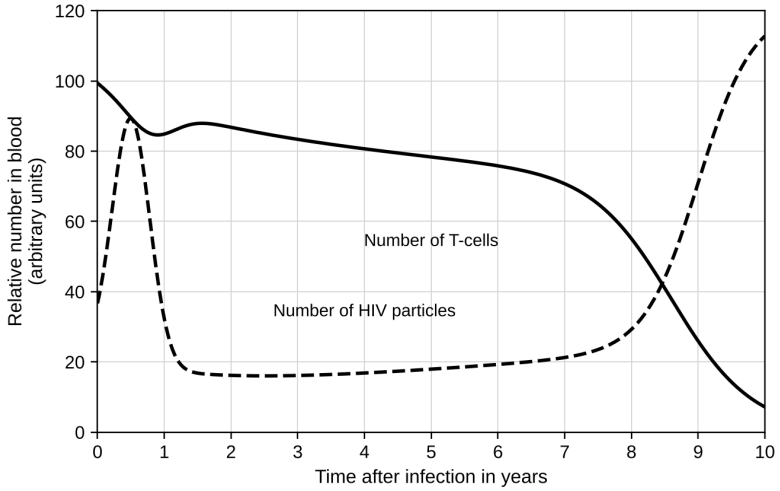
0 3 . 4 Explain why a lower percentage of vaccinated children led to more cases of measles in the unvaccinated children. **[3 marks]**

0 3 . 5 In 2000 the city had 25 000 children. Calculate the number of children who had **not** been vaccinated. **[2 marks]**

Answer = _____

0 3 . 6 Suggest **one** reason why the number of cases fell between 2002 and 2008. **[1 mark]**

0 4 HIV is a virus that infects white blood cells called T-cells. The graph shows what happens to the number of HIV particles and the number of T-cells in the blood of an infected person.



0 4 . 1 Describe what happens to the number of HIV particles in the first 2 years. **[2 marks]**

0 4 . 2 Use the graph to find the number of T-cells at 4 years. **[1 mark]**

0 4 . 3 Describe the pattern shown by the T-cells between 8 and 10 years. **[2 marks]**

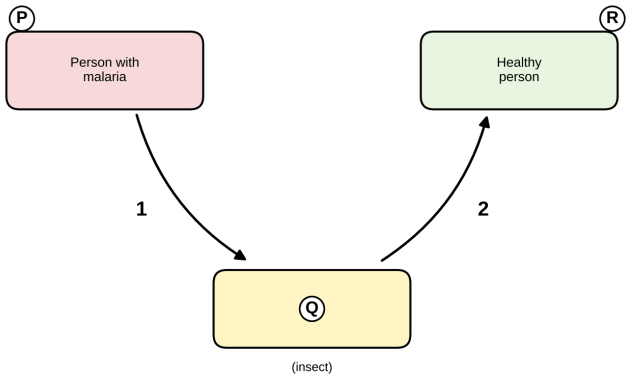
0 4 . 4 Explain why the number of HIV particles rises after 8 years. **[2 marks]**

0 4 . 5 Explain why a person in the late stage of HIV infection is more likely to get other infections. **[2 marks]**

0 4 . 6 Antiretroviral drugs are used to treat HIV. Explain why they do not **cure** the disease. **[1 mark]**

0 4 . 7 Suggest **two** ways in which the spread of HIV can be reduced. **[2 marks]**

0 5 The diagram shows how malaria spreads. Malaria is caused by a protist and is passed on by a vector.



0 5 . 1 What is meant by a **vector**? **[1 mark]**

0 5 . 2 Name the organism labelled **Q**. **[1 mark]**

0 5 . 3 Describe what happens at arrow **1** and arrow **2**. **[2 marks]**

0 5 . 4 Give **three** ways to reduce the spread of malaria. **[3 marks]**

0 5 . 5 Gonorrhoea is a sexually transmitted infection (STI) caused by bacteria. **[2 marks]**
Give **two** symptoms of gonorrhoea.

0 5 . 6 Explain why gonorrhoea is now harder to treat than it was in the past. **[2 marks]**

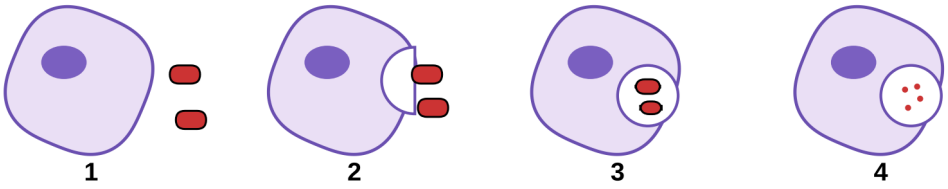
0 5 . 7 Name **one** way of preventing the spread of gonorrhoea. **[1 mark]**

0 6 The human body has several ways of preventing pathogens from entering and of destroying them.

0 6 . 1 Draw **one** line from each body part to its defence against pathogens. **[4 marks]**

Body part	Defence
skin	Cilia and mucus
nose	Hydrochloric acid
trachea	Hairs and mucus
stomach	Barrier / platelets seal cuts

0 6 . 2 The diagrams show a white blood cell destroying bacteria. Describe what is happening in stages **1** to **4**. **[4 marks]**

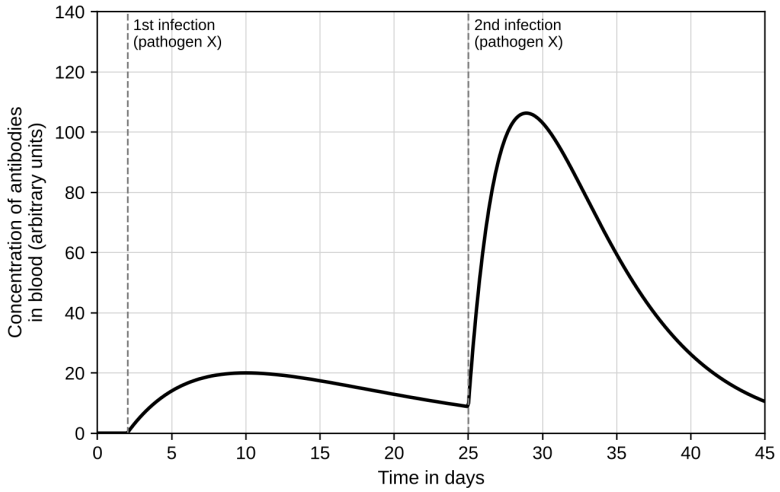


0 6 . 3 Name the process shown in the diagrams. **[1 mark]**

0 6 . 4 Smoking damages the cilia in the trachea. Explain why smokers get more chest infections. **[2 marks]**

0 6 . 5 Platelets are fragments of cells in the blood. Describe how they help to protect the body from pathogens. **[2 marks]**

0 7 A person is infected with a pathogen, **X**, on day 2. The person is infected with the same pathogen again on day 25. The graph shows the concentration of antibodies in the blood.



0 7 . 1 Name the type of white blood cell that produces antibodies. **[1 mark]**

0 7 . 2 Explain how antibodies destroy pathogens. **[3 marks]**

0 7 . 3 Use the graph to find how many days it took after the first infection for the antibody concentration to reach its highest point. **[1 mark]**

0 7 . 4 Describe **two** differences between the response to the first infection and the response to the second infection. **[2 marks]**

0 7 . 5 Explain why the response to the second infection is faster. **[2 marks]**

0 7 . 6 Antitoxins are also made by white blood cells. Describe the function of antitoxins. **[1 mark]**

0 8 Vaccination protects people from some infectious diseases.

0 8 . 1 A vaccine contains dead or inactive pathogens. Explain how a vaccine stops a person getting ill when they are infected with the live pathogen later. **[4 marks]**

0 8 . 2 A trial tested a new vaccine. 5000 people were given the vaccine and 5000 were given no vaccine. After 1 year 40 vaccinated people and 400 unvaccinated people caught the disease.
Calculate how many times more likely an unvaccinated person was to catch the disease. **[3 marks]**

Answer = _____ times

0 8 . 3 Explain what is meant by **herd immunity**. **[2 marks]**

0 8 . 4 Some babies cannot be vaccinated because they are too young. Explain how high vaccination rates in other people still protect these babies. **[2 marks]**

0 8 . 5 Give **one** advantage and **one** disadvantage of vaccination for an individual. **[2 marks]**

0 8 . 6 Why are new vaccines needed each year for flu but not for measles? **[1 mark]**

0 9 This question is about treating infections.

0 9 . 1 Tick **one** box to show which statement about antibiotics is correct. **[1 mark]**

*Tick **one** box.*

- ☐ Antibiotics kill viruses inside the body
- ☐ Antibiotics kill bacteria inside the body without harming human cells
- ☐ Antibiotics treat the symptoms of a cold
- ☐ Antibiotics are painkillers

0 9 . 2 Explain why antibiotics cannot be used to treat measles. **[2 marks]**

0 9 . 3 Painkillers do not kill pathogens. Describe how painkillers help a patient with an infection. **[1 mark]**

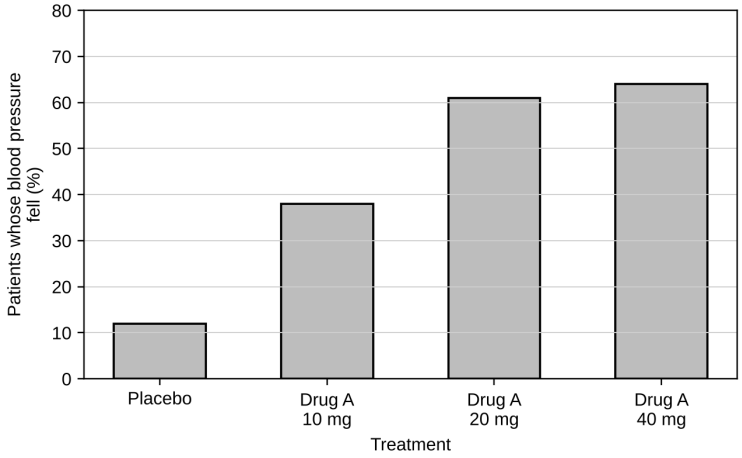
0 9 . 4 Some strains of bacteria are resistant to antibiotics. Explain how antibiotic-resistant strains of bacteria develop. **[4 marks]**

0 9 . 5 A hospital reports a resistant strain called MRSA. Suggest why this is a problem for patients. **[2 marks]**

0 9 . 6 Suggest **two** ways of reducing the number of antibiotic-resistant strains. **[2 marks]**

0 9 . 7 Penicillin was discovered by Alexander Fleming. Describe how it was discovered. **[2 marks]**

1 0 A company develops a new drug, **Drug A**, to reduce high blood pressure. The graph shows the results of a clinical trial. In the trial patients were given a placebo or Drug A.



1 0 . 1 What is a **placebo**? **[1 mark]**

1 0 . 2 Suggest why some patients' blood pressure fell after taking the placebo. **[1 mark]**

1 0 . 3 Describe the effect of the dose of Drug A on the percentage of patients whose blood pressure fell. **[2 marks]**

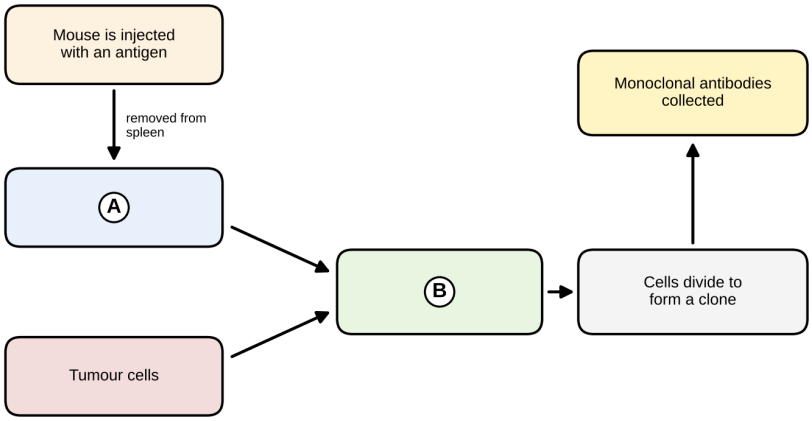
1 0 . 4 The company recommends a dose of 20 mg rather than 40 mg. Suggest **two** reasons. **[2 marks]**

10.5 The trial was **double-blind**. Explain what this means and why it is used. **[2 marks]**

10.6 Before a trial on patients, the drug is tested in the laboratory and on healthy volunteers. Describe **three** stages in testing and developing a new drug. **[3 marks]**

10.7 Why must results of drug trials be checked by other scientists (peer review) before the drug is used? **[1 mark]**

11 Monoclonal antibodies are antibodies that are all identical. The diagram shows how they are produced.



11.1 Name the cells labelled **A** and **B**. **[2 marks]**

1 1 . 2 Explain why the tumour cells are used. **[2 marks]**

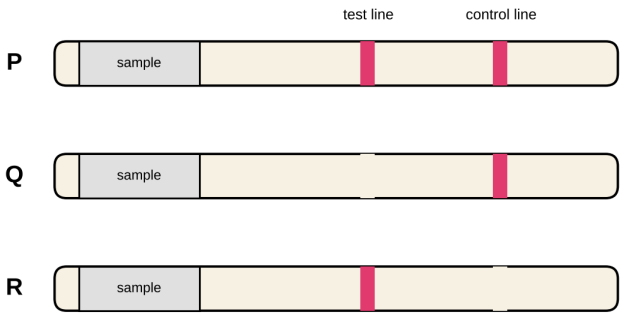
1 1 . 3 Explain why monoclonal antibodies can be used to target a particular chemical in the body. **[2 marks]**

1 1 . 4 Monoclonal antibodies can be used to treat cancer. Describe how. **[3 marks]**

1 1 . 5 Monoclonal antibodies have caused more side effects than expected in some patients. Suggest why this limits their use. **[1 mark]**

1 1 . 6 Give **two** other uses of monoclonal antibodies. **[2 marks]**

1 2 A pregnancy test strip contains monoclonal antibodies. Urine is added at the sample end. The test line contains fixed antibodies to the hormone hCG. The sample pad has antibodies to hCG with a pink dye attached. The control line catches the dye-labelled antibodies. The diagram shows the results for three tests, **P**, **Q** and **R**.



1 2 . 1 Which test, **P**, **Q** or **R**, shows that the woman is pregnant? **[1 mark]**

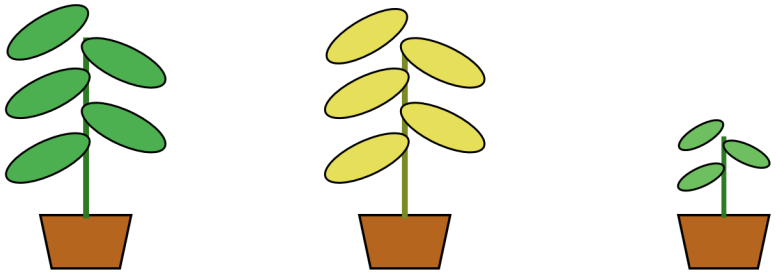
1 2 . 2 Explain how the test line becomes coloured when the woman is pregnant. **[3 marks]**

1 2 . 3 Test **R** is described as invalid. Suggest why. **[2 marks]**

1 2 . 4 Suggest why the test line in test **Q** is not coloured. **[1 mark]**

12.5 Explain why antibodies are used in the pregnancy test and not another type of protein. **[2 marks]**

13 Plants can suffer from diseases and from a lack of mineral ions. The diagram shows three plants, **X**, **Y** and **Z**, of the same species grown in different soils.



13.1 Plant **X** is healthy. Plant **Y** has yellow leaves. Name the mineral ion that plant **Y** is short of and explain why this causes yellow leaves. **[3 marks]**

13.2 Plant **Z** shows stunted growth. Name the mineral ion that plant **Z** is short of. Explain why this stunts growth. **[3 marks]**

13.3 Tobacco mosaic virus (TMV) causes a pattern of discoloured leaves. Explain why TMV reduces the growth of the plant. **[2 marks]**

1 3 . 4 Rose black spot is a fungal disease. Describe **two** ways a gardener can reduce its spread. **[2 marks]**

1 3 . 5 Describe how plants can be tested to identify a particular disease. **[2 marks]**

1 3 . 6 Name **three** signs that a plant is diseased. **[3 marks]**

1 3 . 7 Give **two** physical (mechanical) adaptations that help plants to protect themselves against pests. **[2 marks]**

1 4 Stretch questions (grade 7–9).

1 4 . 1 *Describe how the human body defends itself against pathogens. Include barriers, the roles of white blood cells and how the body responds the second time it meets the same pathogen. **[6 marks]**

1 4 . 2 *A scientist says: 'A new vaccine should be used as soon as it has been shown to be effective in animals.' Evaluate this statement. **[6 marks]**

1 4 . 3 *Explain how the use of antibiotics has led to the development of antibiotic-resistant bacteria and describe how the problem could be reduced. **[6 marks]**

END OF QUESTIONS

MARK SCHEME – Topic 4.3: Infection and response

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	a microorganism that causes disease		1	AO1 4.3.1.1
01.2	B		1	AO2 4.3.1.1
01.3	D		1	AO2 4.3.1.1
01.4	malaria		1	AO1 4.3.1.1
01.5	bacteria divide / reproduce rapidly inside the body ; they produce toxins that damage tissues / cells ;		2	AO1 4.3.1.1
01.6	viruses live and reproduce inside cells ; the cell is damaged / bursts when the virus particles are released ;		2	AO1 4.3.1.1
01.7	3 hours = 9 divisions (180 ÷ 20) ; 2^9 ; 512 ;		3	AO2 4.3.1.1
02.1	Measles: virus ; Salmonella: bacteria ; Malaria: protist ; Malaria: spread by mosquito bites / insect vector ; Rose black spot: fungus ;	One mark for each correct answer in a blank cell. Allow 'Anopheles' / 'vector'.	5	AO1 4.3.1.2
02.2	wash hands / cook food thoroughly ; store raw meat separately / vaccinate poultry ;	Any two sensible hygiene measures.	2	AO1 4.3.1.2
02.3	removes / kills pathogens on the hands ; so they are not transferred to food / surfaces / other people ;		2	AO2 4.3.1.2
02.4	measles: fever / red skin rash ; HIV: flu-like illness at first / later the immune system is damaged ;	Allow: late-stage HIV (AIDS) — immune system too damaged to fight other infections.	2	AO1 4.3.1.3
02.5	it is spread by droplets (airborne) when infected people cough / sneeze ; many people are close together / pupils are not all vaccinated ;		2	AO2 4.3.1.2
03.1	1998: 90 ; 2000: 170 ;		2	AO3 WS 3.1
03.2	170 − 90 = 80 ; 80 ÷ 90 × 100 ; 89 % (88.9) ;		3	AO2 MS 1c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.3	as the percentage vaccinated decreases the number of cases increases (and vice versa) / negative correlation ; e.g. lowest vaccination 80 % in 2000 and highest cases 170 / data quote from the graph ;		2	AO3 4.3.1.4
03.4	vaccinated people are immune and do not pass on the pathogen ; when vaccination falls, herd immunity is lost ; so measles spreads more easily to those who have not been vaccinated ;		3	AO2 4.3.1.4
03.5	100 − 80 = 20 % ; 5000 ;		2	AO2 MS 1c
03.6	more children were vaccinated / vaccination rate rose / herd immunity recovered ;		1	AO3 4.3.1.4
04.1	rises sharply to a peak (about 90) at 0.5 years ; then falls (to about 16) ;		2	AO3 4.3.1.3
04.2	79–83 (arbitrary units)		1	AO3 WS 3.1
04.3	number of T-cells falls rapidly / from about 55 to about 7 ; the number of HIV particles rises at the same time ;		2	AO3 4.3.1.3
04.4	the immune system is weakened / fewer T-cells ; so the body cannot destroy the virus (virus reproduces faster) ;		2	AO2 4.3.1.3
04.5	the immune system is damaged (few T-cells / white blood cells) ; so the body cannot fight other pathogens ;		2	AO2 4.3.1.3
04.6	they slow down / control virus reproduction but the virus remains in the body ;		1	AO2 4.3.1.3
04.7	use condoms ; do not share needles / screen blood donations / treat infected mothers ;		2	AO1 4.3.1.3
05.1	an organism that carries a pathogen (without being ill itself) from one host to another		1	AO1 4.3.1.3
05.2	mosquito		1	AO1 4.3.1.3
05.3	1: mosquito bites the infected person and takes in blood containing the protist ; 2: mosquito bites a healthy person and passes the protist into their blood ;		2	AO2 4.3.1.3
05.4	mosquito nets over beds ; insecticides / insect repellent ; drain stagnant water where mosquitoes breed / introduce predators of larvae ;	Allow: antimalarial drugs; screens on windows.	3	AO1 4.3.1.3
05.5	pain when urinating ; thick yellow or green discharge from the vagina or penis ;		2	AO1 4.3.1.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.6	bacteria have become resistant to penicillin / antibiotics ; so other antibiotics are needed / the infection spreads ;		2	AO2 4.3.1.7
05.7	use a barrier method such as a condom / avoid unprotected sex / screen partners		1	AO1 4.3.1.3
06.1	skin — acts as a barrier / scabs form (platelets) ; nose — hairs and mucus trap particles / pathogens ; trachea — mucus and cilia move trapped pathogens out ; stomach — hydrochloric acid kills pathogens ;	One mark for each correct link.	4	AO1 4.3.1.5
06.2	1: bacteria come near the white blood cell ; 2: the cell engulfs the bacteria (phagocytosis) ; 3: bacteria are enclosed in a vacuole ; 4: enzymes digest / destroy the bacteria ;		4	AO2 4.3.1.5
06.3	phagocytosis		1	AO1 4.3.1.5
06.4	cilia cannot move mucus / pathogens away ; so more pathogens reach the lungs ;		2	AO3 4.3.1.5
06.5	platelets help the blood to clot (form a scab) ; which seals the wound so pathogens cannot enter ;		2	AO1 4.3.1.5
07.1	lymphocyte(s)		1	AO1 4.3.1.5
07.2	each antibody is specific to one antigen (complementary shape) ; antibodies bind to antigens on the pathogen ; which causes the pathogens to clump together / makes them easier for phagocytes to engulf ;		3	AO1 4.3.1.5
07.3	8 days (day 10 − day 2)	Allow 7–9 days.	1	AO3 WS 3.1
07.4	response is faster after the second infection ; higher concentration of antibodies after the second infection ;	Allow quantified: peak about 20 vs about 105; 8 days vs about 4 days.	2	AO3 4.3.1.5
07.5	memory cells (lymphocytes) were produced after the first infection ; they recognise the same antigen and produce antibodies quickly ;		2	AO2 4.3.1.4
07.6	counteract / neutralise toxins released by bacteria		1	AO1 4.3.1.5
08.1	the vaccine contains antigens (from the dead / weakened pathogen) ; white blood cells (lymphocytes) make antibodies against the antigens ; memory cells are produced ; if the live pathogen enters, the antibodies are made quickly / in large amounts so the person does not get ill ;		4	AO2 4.3.1.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
08.2	vaccinated: 40 ÷ 5000 × 100 = 0.8 % ; unvaccinated: 400 ÷ 5000 × 100 = 8 % ; 10 times ;		3	AO2 MS 1c
08.3	if a large proportion of the population is vaccinated ; the pathogen cannot spread easily so unvaccinated people are protected ;		2	AO1 4.3.1.4
08.4	fewer people are infected / carry the pathogen ; so there is less chance of the babies coming into contact with it ;		2	AO2 4.3.1.4
08.5	advantage: protects against serious disease / may prevent death ; disadvantage: possible side effects (e.g. fever, sore arm) / vaccine not 100 % effective ;		2	AO3 4.3.1.4
08.6	the flu virus changes (mutates) so memory cells no longer recognise its antigens ;	Allow 'new strains of flu'.	1	AO3 4.3.1.4
09.1	Antibiotics kill bacteria inside the body without harming human cells		1	AO1 4.3.1.6
09.2	measles is caused by a virus ; viruses live inside cells / antibiotics only kill bacteria ;		2	AO2 4.3.1.6
09.3	they relieve the symptoms (pain) ;	Allow: do not cure the disease.	1	AO1 4.3.1.6
09.4	mutations in the bacterial DNA produce resistance ; antibiotic kills the non-resistant bacteria ; resistant bacteria survive and reproduce ; so the resistant strain becomes more common (natural selection) ;		4	AO2 4.3.1.7
09.5	MRSA cannot be killed by many antibiotics / harder to treat ; it can spread easily between patients in hospital ;		2	AO3 4.3.1.7
09.6	doctors do not prescribe antibiotics for viral infections / unless necessary ; patients finish the whole course of antibiotics ;	Allow: develop new antibiotics; restrict antibiotics in farming; good hygiene in hospitals.	2	AO1 4.3.1.7
09.7	Fleming noticed that mould (Penicillium) had killed the bacteria around it on a Petri dish ; the mould released a chemical that killed bacteria ;		2	AO1 4.3.1.6
10.1	a substance / tablet that looks like the drug but contains no active drug		1	AO1 4.3.1.8
10.2	they expect to get better / the placebo effect ;		1	AO3 4.3.1.8
10.3	as the dose increases the percentage increases ; but there is only a small increase between 20 mg and 40 mg (61 % to 64 %) ;		2	AO3 4.3.1.8
10.4	little extra benefit at 40 mg ; higher dose may cause more side-effects / costs more ;		2	AO3 4.3.1.8

Question	Answers	Extra information	Mark	AO / Spec. Ref.
10.5	neither the patients nor the doctors know who is given the drug or placebo ; to prevent bias / expectation affecting the results ;		2	AO1 4.3.1.8
10.6	tested on cells / tissues / live animals (preclinical) ; clinical trials on healthy volunteers using low doses to check safety ; trials on patients to find the optimum dose / effectiveness ;	Allow peer review / publication of results as an extra stage.	3	AO1 4.3.1.8
10.7	to check the evidence is valid / reliable / not biased ;		1	AO3 4.3.1.8
11.1	A: B lymphocytes (that make the antibody) ; B: hybridoma (cell) ;		2	AO1 4.3.1.9
11.2	tumour cells divide rapidly / indefinitely ; so many identical cells (a clone) can be made to produce a lot of antibody ;		2	AO2 4.3.1.9
11.3	each antibody is specific / binds to one particular antigen ; so they can carry a drug / dye to the cells with the antigen ;		2	AO2 4.3.1.9
11.4	antibodies are made that bind to antigens on cancer cells ; a radioactive substance / toxic drug is attached to the antibody ; so only the cancer cells are killed / healthy cells are not harmed ;		3	AO1 4.3.1.9
11.5	patients become ill / are harmed so they are not used as widely as hoped ;		1	AO3 4.3.1.9
11.6	pregnancy tests (detect hCG) ; to detect / locate pathogens / hormones in diagnosis / in research ;		2	AO1 4.3.1.9
12.1	P		1	AO3 4.3.1.9
12.2	hCG in the urine binds to the dye-labelled antibodies ; the hCG / antibody complex is caught by the fixed antibodies at the test line ; the dye builds up to give a pink line ;		3	AO2 4.3.1.9
12.3	no control line shows ; so the test has not worked properly (dye-labelled antibody not reaching the line / not enough urine) ;		2	AO3 4.3.1.9
12.4	there is no hCG in the urine / the woman is not pregnant ;		1	AO3 4.3.1.9
12.5	antibodies are specific ; only bind to hCG / so other substances in urine do not give a false result ;		2	AO2 4.3.1.9

Question	Answers	Extra information	Mark	AO / Spec. Ref.
13.1	magnesium (ions) ; needed to make chlorophyll ; without chlorophyll leaves turn yellow / cannot photosynthesise well ;		3	AO2 4.3.2.2
13.2	nitrate (ions) ; needed to make amino acids / proteins ; which are needed for growth ;		3	AO2 4.3.2.2
13.3	the discoloured parts have less chlorophyll ; so less photosynthesis / less glucose for growth ;		2	AO2 4.3.2.1
13.4	remove and destroy infected leaves ; use fungicides ;		2	AO1 4.3.2.1
13.5	use a gardening manual / website to match symptoms ; send samples to a laboratory / use testing kits containing monoclonal antibodies ;		2	AO1 4.3.2.1
13.6	stunted growth ; spots on leaves ; areas of decay / malformed stems or leaves / discolouration / presence of pests ;		3	AO1 4.3.2.1
13.7	thorns / hairs ; leaves that droop or curl when touched / mimicry / bark / waxy cuticle / cellulose cell walls ;	Any two. Chemical defences (e.g. poisons, antibacterial chemicals) are credited in other questions.	2	AO1 4.3.2.3
14.1	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • barriers: skin, mucus and cilia in the nose and trachea, hydrochloric acid in the stomach • if pathogens enter, white blood cells engulf and digest them (phagocytosis) • lymphocytes make antibodies specific to the antigens on the pathogen • antibodies make pathogens clump / easier to destroy; antitoxins neutralise toxins • memory cells are made which recognise the same antigen • second time: faster and larger production of antibodies, so the person does not become ill		6	AO1 4.3.1.5

Question	Answers	Extra information	Mark	AO / Spec. Ref.
14.2	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • animal tests do not always show what happens in humans; humans might react differently • clinical trials in healthy volunteers check safety at low doses • trials in patients find the best dose and effectiveness; placebo and double-blind trials reduce bias • peer review checks results are valid before the vaccine is used • benefits of early use: protects people sooner; could save lives in an outbreak • risks: unknown side-effects; ethical issues; balanced conclusion: safety testing is needed before widespread use		6	AO3 4.3.1.8
14.3	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • bacteria vary: random mutations produce some resistant individuals • antibiotics kill the non-resistant bacteria; resistant bacteria survive • survivors reproduce and pass on the resistance gene; resistant strain becomes more common (natural selection) • overuse (e.g. for viral infections) increases selection pressure; not finishing the course leaves resistant survivors • reduce: prescribe only when needed; complete the course; avoid antibiotics in farming • develop new antibiotics; hygiene in hospitals to prevent spread (e.g. MRSA)		6	AO2 4.3.1.7

Where marks are lost in Topic 4.3

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Pathogens:** bacteria make toxins and divide fast; viruses live and reproduce inside cells; protists (malaria) need a vector; fungi (rose black spot) spread by spores.
- ☐ **Spread:** always link the disease to the method (droplets, contaminated food, sexual contact, vector bite) and say how it is stopped.
- ☐ **Immune response:** phagocytes engulf, lymphocytes make antibodies (specific to the antigen), antitoxins neutralise toxins. Second infection = faster, higher, because of memory cells.
- ☐ **Vaccines:** contain dead or inactive pathogen (antigens) → antibodies + memory cells. Herd immunity protects those who cannot be vaccinated.
- ☐ **Antibiotics:** kill bacteria, not viruses. Resistance comes from mutation + natural selection — never say 'the bacteria got used to it'.
- ☐ **Drug testing:** cells and animals → healthy volunteers → patients. Use a placebo and double-blind trials; peer review before use.
- ☐ **Percentage change = (change ÷ original) × 100.** Show the working; use the original value as the divisor.
- ☐ **Monoclonal antibodies:** B lymphocyte + tumour cell = hybridoma. Specific to one antigen, so they can target cancer cells or detect hCG.
- ☐ **Plant ions:** nitrate → amino acids / proteins (stunted growth); magnesium → chlorophyll (yellow leaves). Name the ion AND the reason.